Knowledge HUB – Book Sphere – Day-wise Micro Execution Plan

(Backend First | IntelliJ Driven | Feature-Based)

PHASE 0 – Environment + Setup (Day 1–2)

Day 1 – IntelliJ + Project Initialization

Task 1.1 – IntelliJ Setup

What to Do:

- Open IntelliJ IDEA Ultimate
- Install plugins:
    - Lombok
    - Spring Boot Assistant
- Enable annotation processing

What to Learn:

- Project navigation (Project, Structure, Search Everywhere)
- Run configurations

What to Test:

- Create a sample Java class and run

Task 1.2 – Create Spring Boot Project

What to Do:

- Use Spring Initializr inside IntelliJ
- Add dependencies:
    - Spring Web
    - Spring Data JPA
    - MySQL Driver
    - Lombok
    - Validation

What to Test:

- Application runs successfully

Git Commit:

feat: initialize spring boot project

Task 1.3 – Basic API Test

What to Do:

- Create /hello endpoint

What to Test:

- Hit endpoint via browser/Postman

Day 2 – Project Structure + Git

Task 2.1 – Clean Package Structure

What to Do:
Create packages:

controller

service

repository

entity

dto

config

security

exception

util

Task 2.2 – Git Setup

What to Do:

- Create repo: booksphere-lms
- Add .gitignore
- Push code

Git Commit:

chore: project structure setup

Task 2.3 – IntelliJ Debug Practice

What to Do:

- Add breakpoint in controller
- Run in debug mode

What to Learn:

- Step into / step over

PHASE 1 – Database + Base Design (Day 3–5)

Day 3 – Database Setup

Task 3.1 – MySQL Setup

What to Do:

- Install MySQL
- Create DB: library_db

Task 3.2 – Configure Application

What to Do:

- Add DB config in application.properties

Task 3.3 – IntelliJ DB Tool

What to Do:

- Connect DB inside IntelliJ

What to Learn:

- Run queries directly from IDE

Day 4 – Base Entity

Task 4.1 – Create BaseEntity

Fields:

- id
- createdAt
- updatedAt

Task 4.2 – Enable Auditing

What to Learn:

- @MappedSuperclass
- @EntityListeners

Task 4.3 – Test Persistence

What to Do:

- Create dummy entity
- Save to DB

Day 5 – ER Design + Planning

Task 5.1 – Identify Entities

- User
- Role
- Book
- Genre

- Transaction
- Request

Task 5.2 – Define Relationships

What to Learn:

- OneToMany
- ManyToOne

Task 5.3 – Write Notes

What to Do:

- Document schema in /docs

PHASE 2 – Authentication Module (Day 6–12)

Day 6 – User + Role Entity

- Create User entity
- Create Role entity
- Map relationship

Day 7 – Repository Layer

- Create UserRepository
- Create RoleRepository
- Test CRUD

Day 8 – Registration API

- Create DTO
- Create Service
- Create Controller

Test:

- Register user via Postman

Day 9 – Password Encryption

- Configure BCrypt

Test:

- Password stored encrypted

Day 10 – Login API

- Implement authentication logic

Test:

- Login works

Day 11 – JWT Setup

- Create JWT util
- Generate token

Day 12 – Security Config

- Configure Spring Security
- Secure endpoints
- Add role-based access

Day 12 (Part 2) – OTP Email

- Setup SMTP
- Send OTP

PHASE 3 – Book Module (Day 13–20)

Day 13 – Book Entity

- Fields:
    - title
    - author
    - genre
    - totalCopies
    - availableCopies
    - location

Day 14 – Genre Entity

- Create Genre table
- Map with Book

Day 15 – Book Repository

- Add custom queries

Day 16 – Add Book API

- Controller + Service

Day 17 – Update/Delete Book

- Implement APIs

Day 18 – Search API

- Search by title, author

Day 19 – Pagination

- Pageable implementation

Day 20 – Availability Logic

- Update availableCopies

PHASE 4 – Transaction Module (Day 21–28)

Day 21 – Transaction Entity

- Fields:
    - issueDate
    - returnDate
    - dueDate

Day 22 – Issue Book API

- Reduce availableCopies

Day 23 – Return Book API

- Increase availableCopies

Day 24 – Due Date Logic

- Add 12-day rule

Day 25 – Fine Calculation

- Implement fine logic

Day 26 – Borrow History API

Day 27 – Active Loans API

Day 28 – Edge Cases

- Prevent invalid issue
- Handle exceptions

PHASE 5 – Notification Module (Day 29–32)

Day 29 – Email Service

Day 30 – Issue Email

Day 31 – Reminder Scheduler

Day 32 – Overdue Notification

PHASE 6 – Request Module (Day 33–35)

Day 33 – Request Entity

Day 34 – Request API

Day 35 – Admin View Requests

PHASE 7 – Redis (Day 36–38)

Day 36 – Redis Setup

Day 37 – Cache Book List

Day 38 – Cache Search

PHASE 8 – AWS S3 (Day 39–41)

Day 39 – S3 Setup

Day 40 – Upload API

Day 41 – Download API

PHASE 9 – Frontend (Day 42–55)

- Login/Register UI
- Book UI
- Dashboard
- Admin panel

PHASE 10 – Testing + Deployment (Day 56–65)

Day 56–60

- Full API testing
- Fix bugs

Day 61–65

- Deployment
- Final cleanup
- README update

DAILY RULE (VERY IMPORTANT)

Every single day you must:

1. Code
2. Test (Postman)
3. Debug (IntelliJ)
4. Refactor
5. Commit